CO224 Computer Architecture - 2020

Department of Computer Engineering

Lab 4.5 - Building a Simple Processor Extended ISA

There's a Lab 4.5?!? Well, only if you are up for the challenge, extend your processor to support two or more of the following instructions for maximum of **bonus 20 marks**:

`mult 4 1 2` (multiply value in register 1 by value in register 2, and place the result in register 4)
`sll 4 1 0x02` (apply logical shift left 2 times on value in register 1, and place the result in register 4)
`srl 4 1 0x02` (apply logical shift right 2 times on value in register 1, and place the result in register 4)
`sra 4 1 0x02` (apply arithmetic shift right 2 times on value in register 1, and place the result in register 4)
`ror 4 1 0x02` (apply rotate right 2 times on value in register 1, and place the result in register 4)
`bne 0x02 1 2` (if values in registers 1 and 2 are not equal, branch 2 instructions forward)

Note that you should use the same 3-bit `ALUOP` signal as in the previous part. Since you can only have 8 functional units inside the ALU with the 3-bit ALUOP control signal, you need to find a way for the new instructions to share functional units (*hint: can* `sll` *and* `srl` *share a single functional unit?*). You are discouraged to use additional control signals, unless absolutely necessary.

You must make reasonable assumptions for the latencies of any added hardware, and make sure that the new instructions can still complete within one clock cycle (8 time units).

As this is a bonus exercise, you must implement any new functional unit modules without using simple operations (e.g. using operator << for left shifting will not be acceptable).

You must provide a clear description of the instruction encodings, assigned opcodes, timing, and changes made to the datapath+control as a separate report. Two instructions correctly implemented with proper documentation will earn you 4 bonus marks. Thereafter, you can earn 4 more marks per additional instruction with proper documentation. Submit a compressed file ***groupXX_lab4_5.zip*** containing all files of your design along with a testbench and your report.

Have fun coding. May the force be with you!